Developments in PMT Waveform Simulation



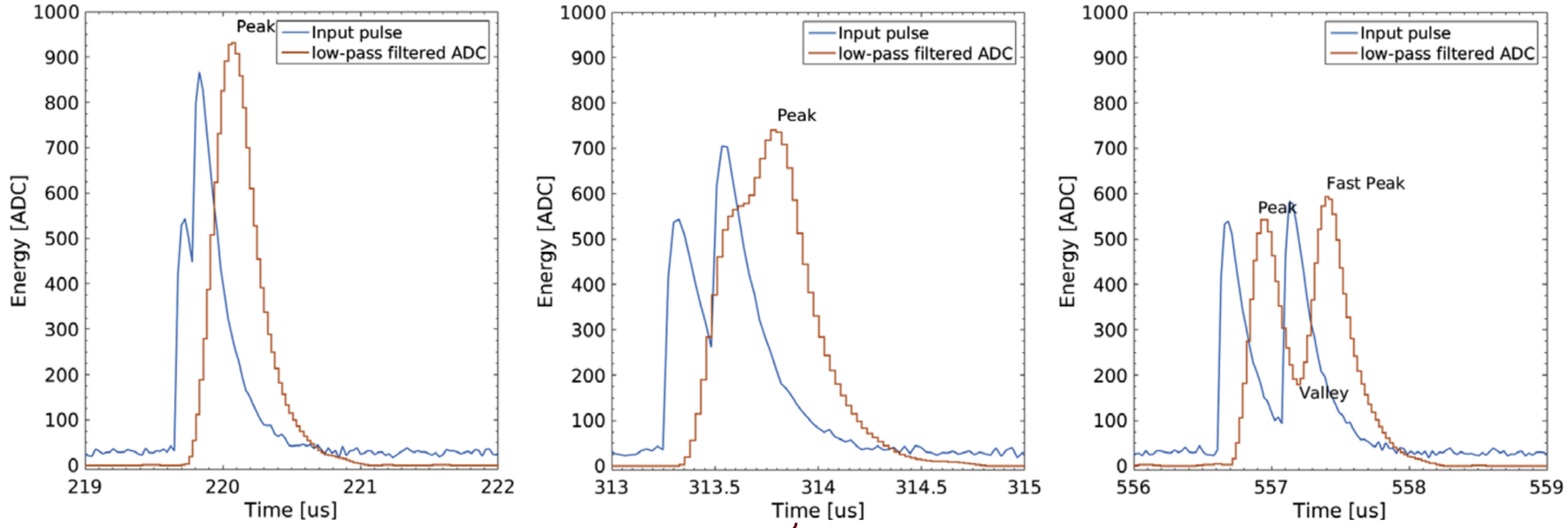
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Goal

- For MC to look as much like data as possible
  - We can trust it more
  - MC needs to reflect the nuance of real data
  - Ideally we'd be able to use the exact same ToolChains on both

Simulated PMT pulses (https://doi.org/10.1007/s11214-018-0573-7)



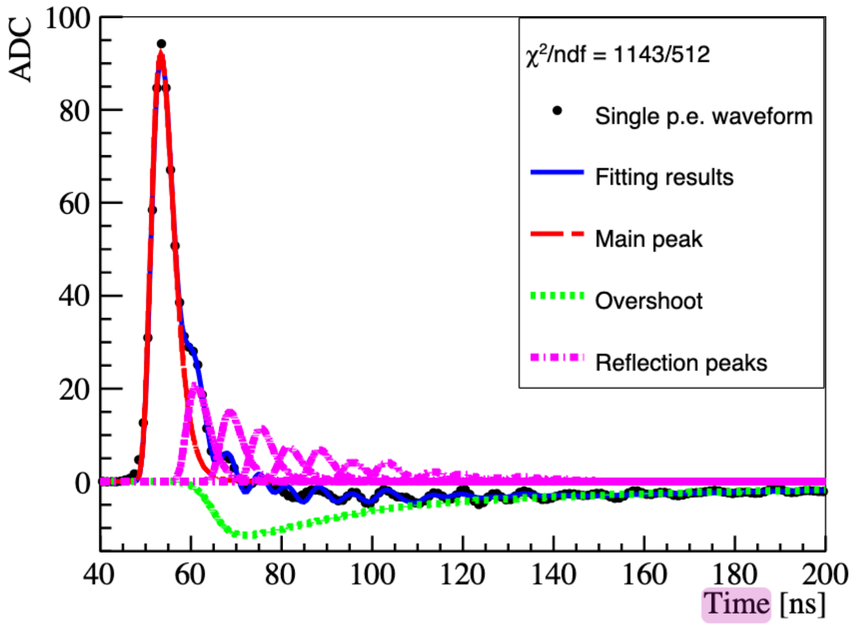
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Waveform Fit

$$U_{peak}(t) = U^0_{peak} \times exp(-\frac{1}{2}(\frac{ln((t-t_0)/\tau)}{\sigma})^2) \quad (2)$$

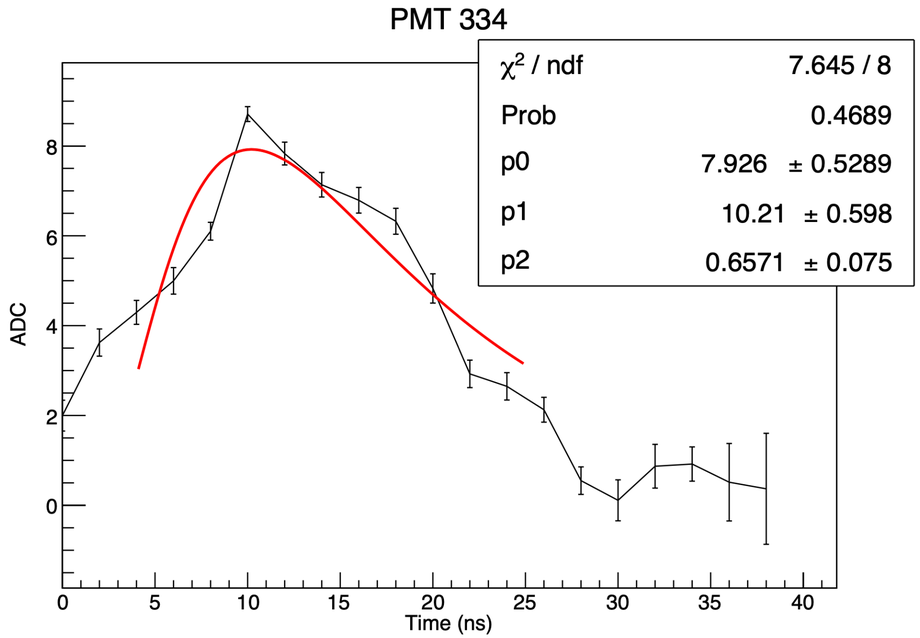
$$U_{os}(t) = U_{peak}(t) \times [U^{Fast}_{os}/(e^{\frac{t_0-t}{10ns}}+1) \times exp(-t/\tau_{fast}) + U^{Slow}_{os}/(e^{\frac{t_0-t}{10ns}}+1) \times exp(-t/\tau_{slow}) + U^{Gaus}_{os} \times exp(-\frac{1}{2}(\frac{t-t_0}{\sigma_{os}})^2)] \quad (3)$$

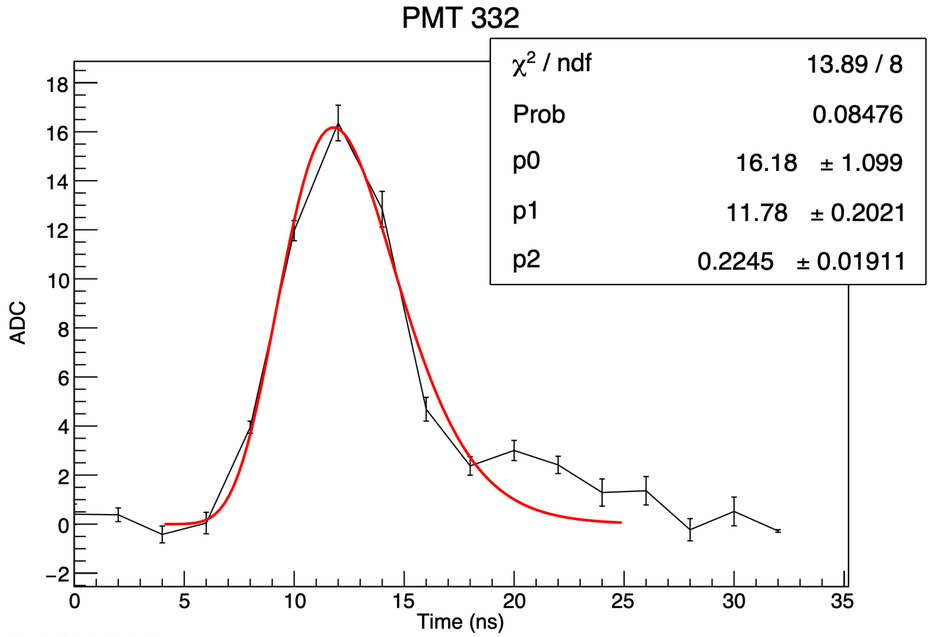
- Following Daya Bay's paper
  - Collect many SPE waveforms for each PMT
  - Average them out and perform the fit
  - Apply the function to each true photon and sum waveforms accordingly
  - Ignoring the overshoot and "reflections" for now
- Looked at laser runs:
  - 4691, 4692, 4694, and 4695
  - Keeping the raw waveform, but subtracting the baseline
  - Waveforms must be SPE-like (integrated Q: 0.5-1.5)

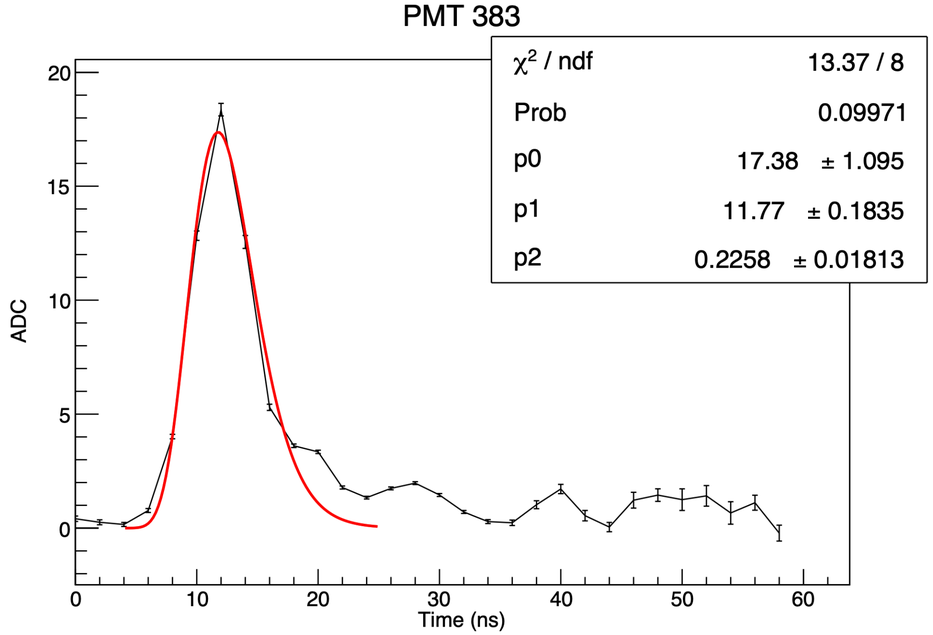


Fit Examples

- Most fits look fairly reasonable
  - p0 encodes the PMT's gain
- Some look quite bad
  - Use some average values





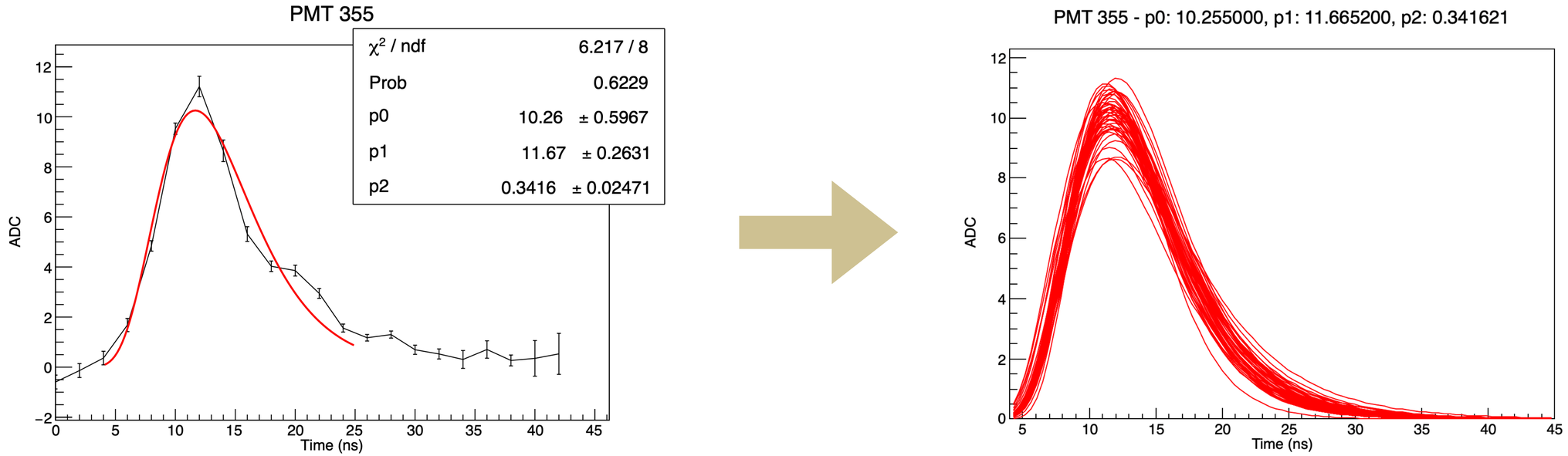


Waveform Simulation

- LoadWCSim saves a map of PMT IDs to a vector of MCHits
  - We turn off hit time smearing in order to use the true hit time
- Each MCHit gets an individual simulated waveform
- Overlapping waveforms are added together
- Random noise and a random pedestal is added
  - Pedestal is drawn uniformly between 300 and 350 ADC
  - Noise sigma is sampled from a Gaussian, then noise is sampled from a new Gaussian with that std dev.
- Full extended readout window (34,993 clock ticks) is simulated

Randomizing the fit parameters

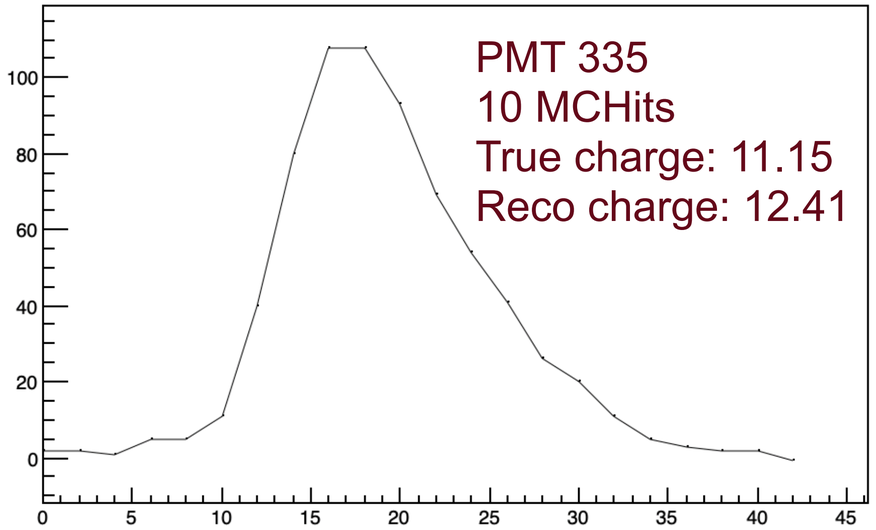
- The fit parameters have associated error bars
  - Can sample each parameter in a way that preserves the fitted covariance
  - Do this for each MCHit to capture random variation in PMT response

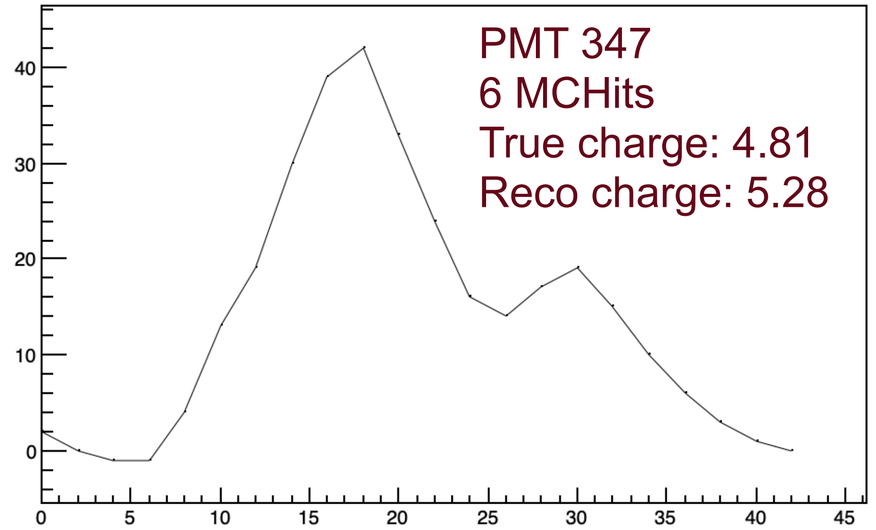


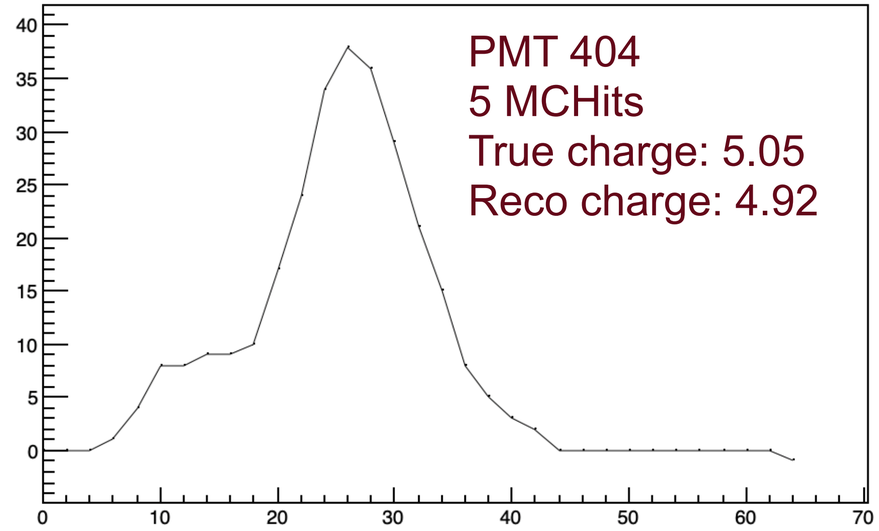
A few more details

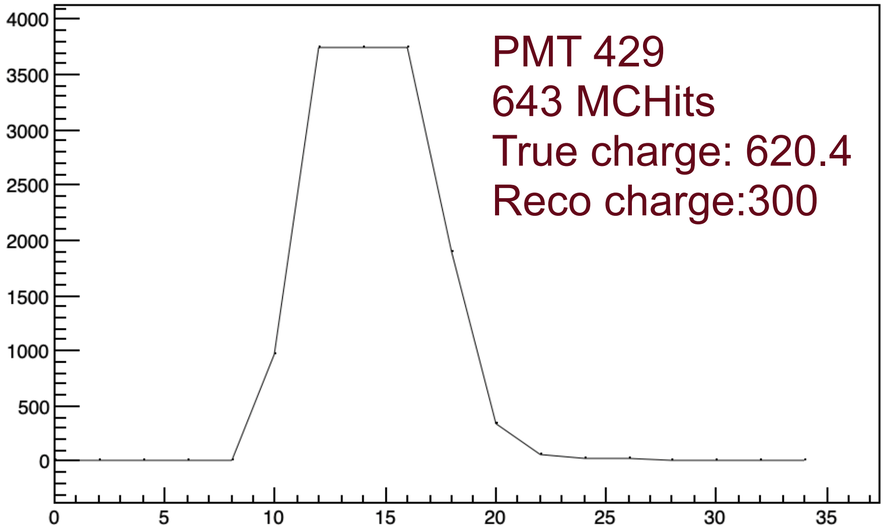
- Need to associate MC truth information with the eventual pulses
- Created a new MCWaveform class that inherits from Waveform
  - Contains a vector of parent IDs (for integration with BackTracker)
  - Allows for simple integration into the PhaseIIADCHitFinder
- CalibratedADCWaveforms are also generated
  - *Could* set it up to pass through the PhaseIIADCCalibrator, but that's a bit of a pain
  - Instead, just use the exact baseline and noise sigma that was generated and scale ADC_TO_VOLT
  - The waveform fits were performed on raw SPE waveforms so gains are implicitly applied

The Results


PMT 335
10 MCHits
True charge: 11.15
Reco charge: 12.41


PMT 347
6 MCHits
True charge: 4.81
Reco charge: 5.28


PMT 404
5 MCHits
True charge: 5.05
Reco charge: 4.92


PMT 429
643 MCHits
True charge: 620.4
Reco charge:300

Summary

- We have developed a PMT waveform simulation
  - Captures features of near-in-time hits
  - Produces PMT saturation
- Integrates with the PhaseIIADCHitFinder
  - Thus, can be treated like data in downstream tools (ie. ClusterFinder)
- Decent agreement between true and reco charge

And one more thing

- Found a bug in PhaseIIADCCalibrator
  - Waveform baselines are determined using the ze3ra method
  - Specifically, ze3ra_multi, which allows for mutiple baseline regions to be used
  - The "SigmaBaseline" (ie. standard deviation of noise around the baseline) is incorrect in this method
  - It was taking the variance in the mean of the multiple baselines
    - If there is only one baseline found, then the variance is 0...
  - To properly combine std devs you need to average the variances from each baseline